



MINHAJ ISLAMIC MODEL SCHOOL HAZRO
FIRST TERM EXAMINATION (2026)

Class: 10th Subject: Chemistry Total Marks: 50

Roll No: _____ Student Name: _____ Time Allowed: 2 Hours

SECTION A: OBJECTIVE TYPE (10 Marks)

Q1. Choose the correct option for each question:

(i) According to kinetic theory, the basic difference between solid, liquid and gas is due to:

[A] the difference in the movements of the particles. [B] the chemical properties of the particles. [C] the size of the particles [D] the shape of the particles

(ii) Upon heating the rate of evaporation:

[A] decreases [B] increases [C] remains the same [D] initially decreases and then increases upon further heating

(iii) Cooling the vapours of some gases change them directly into solid state. This phenomenon is called:

[A] evaporation [B] condensation [C] sublimation [D] deposition

(iv) Physical state in which the particles possess the maximum energy:

[A] Solid [B] Liquid [C] Gaseous [D] Vapour

(v) Which gas will diffuse at the fastest rate?

[A] HCl [B] SO₂ [C] H₂S [D] CO₂

(vi) When natural gas burns in air, which component is the limiting reactant?

[A] Both components [B] Neither component is a limiting reactant [C] Air [D] Natural gas

(vii) Which law states that the volume of a given mass of gas is directly proportional to temperature?

[A] Boyle's Law [B] Charles's Law [C] Graham's Law [D] Avogadro's Law

(viii) What does percentage purity indicate in a chemical sample?

[A] The degree to which a substance is free from impurities [B] The total mass of the container holding the sample [C] The amount of pure compound present in an impure sample [D] The rate of evaporation of solution

(ix) What does the molecular formula of a substance represent?

[A] The simplest ratio of atoms, [B] The actual number of atoms present in a molecule [C] Only the percentage composition [D] The atomic mass of an element.

(x) What is the empirical formula of hydrogen peroxide?

[A] H₂O₂ [B] HO [C] H₂O [D] OH₂

SECTION B: SHORT ANSWER QUESTIONS (10 × 3 = 30 Marks)

Q2. From where does the energy come when a liquid evaporates?

Q3. Is condensation an endothermic process?

Q4. Why does the temperature remain constant during a phase change?

Q5. Can you change the temperature at which water boils in an experiment in your lab?

Q6. Differentiate between evaporation and boiling.

Q7. How does the concept of molar volume useful for us?

Q8. Why is percentage yield important?

Q9. What is the difference between an empirical formula and a molecular formula? Give one example.

Q10. How would you identify the limiting reactant in a chemical reaction?

Q11. Define concentration and give its unit

SECTION C: DESCRIPTIVE / LONG QUESTIONS ($2 \times 5 = 10$ Marks)

Q12. What is Limiting Reactant? How to identify limiting reactant? Discuss the steps

Q13. What is Sublimation. Give the application of sublimation

GOOD LUCK ☺.